Shiva Sharma

R&D; Software Engineer • C / C++ • Optical Networking

(+91) 7404671855 | shivasharma9896@gmail.com | linkedin.com/in/shivasharma9896 | Bengaluru, India

PROFESSIONAL SUMMARY

Embedded software engineer with ~3 years of hands-on C/C++ development experience in carrier-class optical networking and telecom systems on Linux. Proven ability to deliver production-quality features end-to-end — from system design through integration, testing, and release — within cross-functional agile teams. Deep domain expertise in OTN control plane, GPON/XGSPON/NGPON, OMCI, QoS, GCC, and inter-chassis networking. Strong focus on code quality, rigorous code reviews, and continuous improvement through automated testing.

WORK EXPERIENCE

Engineer | Tejas Networks

May 2025 – Present

- Designing and implementing C/C++ control plane software for Optical Transport Network (OTN) devices in a Linux embedded environment, improving network reliability and efficiency.
- Developed and optimized Generic Control Channel (GCC) handling for Line and Client Ports, enabling reliable embedded management communication across OTN links.
- Implemented inter-card and inter-chassis connectivity solutions, enabling seamless communication across distributed hardware in multi-card systems.
- Developing and enhancing Wi-Fi 6 ONT Mesh technology, optimizing mesh networking performance across product portfolio.

R&D; Engineer | Nokia

Aug 2023 – Apr 2025

- Developed embedded C/C++ software for SFU ONT devices, specializing in QoS data packet management and OMCI-level communication across GPON, XGSPON, and NGPON.
- Delivered major new features end-to-end — from design and implementation to customer release — significantly enhancing product functionality.
- Led design pattern and technology selection for new modules, navigating chipset compatibility, vendor limitations, and integration with existing systems in a multi-threaded Linux environment.
- Developed scalable, maintainable solutions under real-world constraints of various chipsets and third-party modules.
- Conducted rigorous code reviews and maintained automated test suites, upholding high standards of code quality and system stability.
- Collaborated closely with hardware engineers, software teams, and third-party vendors to ensure seamless system integration.

Student Trainee (Intern) | Nokia Solutions and Networks

10 months

- Worked on the L2 forwarding team for SFU ONTs, developing and maintaining embedded software for Optical Network Terminal devices.
- Enhanced ONT device performance through continuous software updates and troubleshooting in a Linux environment.
- Received a letter of appreciation; performance led to conversion to a full-time R&D; Engineer role.

EDUCATION

Master of Computer Applications (MCA) | **Vellore Institute of Technology** *2023*

Bachelor of Computer Applications (BCA) | **Gujarat University / St. Xavier's College** *2021*

SKILLS

Programming Languages

C • C++ (Advanced / Multi-threaded) • SQL

Domains & Technologies

OTN Control Plane • GCC (Line & Client Ports) • GPON / XGSPON / NGPON • OMCI • QoS • Inter-card / Inter-chassis Networking • Wi-Fi 6 Mesh • Embedded Systems

Tools & Methodologies

Linux • Agile / TDD • CI/CD Pipelines • Code Review • Automated Testing • Git

CERTIFICATIONS

- SQL — HackerRank
- C++ — Coding Blocks
- Introduction to Embedded Systems — University of Colorado Boulder (Pursuing)